

DANG THAI GIA THUAN

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EDUCATION

VNU-HCM University of Science

2017 – 2022

Bachelor of Computer Science

Department of Information Technology

GPA: 3.0/4.0

Relevant coursework:

- Probability and Mathematical Statistics : 4.0/4.0
- Deep learning for Data Science : 3.0/4.0

CAREER OBJECTIVE

Passionate about advancing artificial intelligence through innovative machine learning and deep learning solutions, particularly in computer vision. Eager to contribute to AI development as an AI Engineer, leveraging expertise in model training, optimization, and data processing to build impactful solutions. Seeking to collaborate with dynamic teams to drive cutting-edge AI applications for real-world challenges.

WORK EXPERIENCE

Quality Control (QC), KBM Company

October 2022 – Present

Binh Thanh District, Ho Chi Minh City

- Evaluated images of Suntory PepsiCo beverage product displays in stores and supermarkets, performing quality assurance tasks.
- Classified images as pass/fail and assessed product availability (in-stock or out-of-stock) to ensure compliance with display standards.
- Contributed to maintaining high-quality visual merchandising standards through detailed image analysis.

AI Engineer, Eimage Development

November 2021 – September 2022

Tan Binh District, Ho Chi Minh City

- Developed classification models using EfficientNet and MobileNet for Pass/Fail image detection, achieving high accuracy in quality control tasks.
- Performed data annotation and preprocessing for computer vision projects, ensuring high-quality training datasets using tools like GLabeller.
- Collaborated with the team to optimize model performance and streamline data pipelines.

- Tools: Python, PyTorch, OpenCV, GLabeler.

PROJECTS

Drink detection system [GitHub]

April 2025 – May 2025

- Developed a drink detection system using state-of-the-art object detection models (e.g. YOLOv8, v9, v10) on a custom drinks dataset, achieving high precision with YOLOv8 version at 80 epochs.
- Manually collected over 400 images of drink data, annotated bounding boxes using LabelMe tool, and implemented preprocessing scripts for processing training data.
- Built a Gradio-based web app for real-time drink detection, supporting image uploads and displaying product counts (e.g., "Sting Vàng Lon 330ml").
- Tools: Python, Ultralytics, Gradio, OpenCV, LabelMe.

Invoice information extraction [GitHub]

January 2025 – March 2025

- Developed an OCR pipeline to extract product names and prices from invoices using VietOCR library and EraX-VL-7B-V1 VLM-based model, leveraging pre-trained models for text detection and information extraction.
- Preprocessed invoice images with OpenCV and rembg package, applying background subtraction and perspective correction to enhance text visibility.
- Performed text detection using CRAFT algorithm and custom post-processing to organize OCR results effectively.
- Utilized prompt-engineering for VLM models to extract required information from invoices.
- Tools: Python, VietOCR, OpenCV, PyTorch, Huggingface Transformers.

Image style transformation [GitHub]

May 2024 – June 2024

- Developed a pipeline using Stable Diffusion XL and SOTA InstantID model to transform human portraits into anime and cartoon styles while preserving identity and pose.
- Integrated ControlNet with pose and canny edge detection for enhanced control over image generation, utilizing face embeddings from InsightFace.
- Implemented BLIP for automated caption generation and conducted hyperparameter surveys to optimize guidance scale and strength for style consistency.
- Tools: Python, PyTorch, Diffusers, ControlNet, InsightFace, BLIP.

Parallel programming final project [GitHub]

January 2021

- Optimized Seam Carving using parallel programming in Cuda.
- Input: Image RGB, Output: Image resized (in width) without distorting important objects.
- Language/Framework: CUDA, Python.

Machine learning final project [GitHub]

August 2020

- Used SVM (Support Vector Machine) model for handwriting recognition on MNIST dataset.
- Trained Linear Kernel models with spread C values to find the C value with the smallest error. Running the SVM (Linear kernel) algorithm with the C value around the best C value found.
- Visualized model performance across parameters to identify the optimal C value. Analyzed performance metrics to evaluate and select the optimal C-parameter.

- Language/Framework/Tool: Python, Scikit-learn.

LICENSES & CERTIFICATIONS

- The Computer Vision & Applications course by VietAI [CERTIFICATE]
- The Pre Machine Learning course by VietAI [CERTIFICATE]
- Convolutional Neural Networks in TensorFlow [CERTIFICATE]
- Natural Language Processing in TensorFlow [CERTIFICATE]
- DeepLearning.AI TensorFlow Developer [CERTIFICATE]

SKILLS

- Programming Languages: Python, C/C++, SQL
- Frameworks: PyTorch, TensorFlow, Keras, OpenCV, YOLO
- Tools: GitHub, LabelMe, Jupyter Notebook
- Platforms: Linux, Windows
- Language: English
- Soft Skills: Attention to detail, meticulousness, teamwork, communication, presentation

ACTIVITIES

"Spring volunteer" campaign

January 2018, January 2020